

laborate with in the third approach is decided in the pre stages, whereas in the second approach a creator starts looking at possible collaborations after knowing what kind of solution is the client looking for.

“I see this as an opportunity which still is a blank slate. If I am aggregating a large sum of data from a large manufacturer, each product queue would have a sensor. I, as a manufacturer, would not sell it to my customer but I would be needing a distributor to sell that data. This entire process is painful because it involves costs at every stage due to the presence of people at every stage. If you can charge for that data, the benefits include better time to market. The data that you have given to an IoT company is now giving you money in an iterative fashion,” feels Trushit.

However, any end-to-end solution cannot be completed unless there are multiple parties involved in creating and deploying that solution. Some examples of teams that are required to be involved in any end-to-end solution are cloud teams, developers’ teams, applications providers, edge computing teams, and more. All these teams, in a lot of cases, can either be from the same vendor or from different vendors.

Since, there is an ecosystem working on the same case, it can end up making win-win for everybody. Such collaborations can not only help democratise the data but it can also open doors for providers to work on definite solutions instead of forcing them to work on everything on their own. The same can also help bring down the time required to develop solutions as different parties are working on different aspects of the same.

“The only challenge is people will have to change the way they pitch solutions. Instead of saying I provide this, people will have to say I am responsible for this. Providing and moving out doesn’t mean you are concerned about the client,” says Anand.

The question of end-to-end solution is also more pertinent from the buyer’s side as it is he who is going to be spending on getting the solution deployed. Hence, the benefits of the same should be clearly visible to the



*Smart electricity meters are in fact the biggest example of how big the IoT vertical is getting*

customer. Such solutions, as per Amit, should give more clarity on the outcome to the consumer. Most customers are usually focused on what they need and talk less about which technologies can help them reach there.

“It, as an end-to-end solution, is better if it’s given to one service provider. A small problem in the solution can cause a mad rush and blame game among all the providers in the ecosystem,” feels Amit. He adds, “An end-to-end solution from a single source reduces risks and makes the single party responsible for everything that is part of that solution. Moreover, innovation is much faster with integrated IoT.”

From the solution provider’s point of view, the biggest battle in this journey usually start when different pieces of the equation do not fit together. There could be times when the hardware available does not mix well with the infrastructure of a company, or there can be instances where software-hardware integration become difficult. These instances usually leave creators and service providers with just one option, and that is to start everything afresh.

For example, a manufacturer working on SMT lines may find sensors not good enough to gather data that he really wants. Similarly, there could be manufacturing units where heat generated is not allowing the sensors

to function properly. Here, better sensors (the new ones) will cost more and will require more time to be proven to work efficiently.

“I think the issues come when you are building a solution and using someone else’s product. There’s only a certain amount of control that you have over what you can do with what’s available. So, as a provider you must always be clear on what you can build from scratch and what you need to outsource to build from scratch. However, at the end of the day, you as a company should be accountable for the solution that you are providing. This should be irrespective of the number of vendors working with you,” says Abhishek.

He adds, “You should not go to a client and tell him the problem was caused due to a certain fault in a certain part of the process provided by a vendor of yours. No client would like to hear that someone else is accountable when you were the one who pitched and provided the solution. Clients would not like to run to third-party vendors in any case!”

An IoT solution makes absolute sense when the responsibility for it to work flawlessly is taken absolutely by one party. There is no way a client would expect data management from someone else and making the sensors function the way they were supposed to from someone else. While there can be multiple vendors responsible for creating an end-to-end IoT solution, the responsibility for making sure that the solution works as per the standards set should always be on the shoulders of one party. There’s a saying that goes well in such scenario: Too many cooks spoil the broth!

## The startup part

Given the present circumstances, there is a clear mandate that the health and medical industry is currently leading in terms of IoT adoption. Hospitals, primary care nursing homes, pharmaceutical companies among other verticals of the health and medical industry are trying to find and imple-